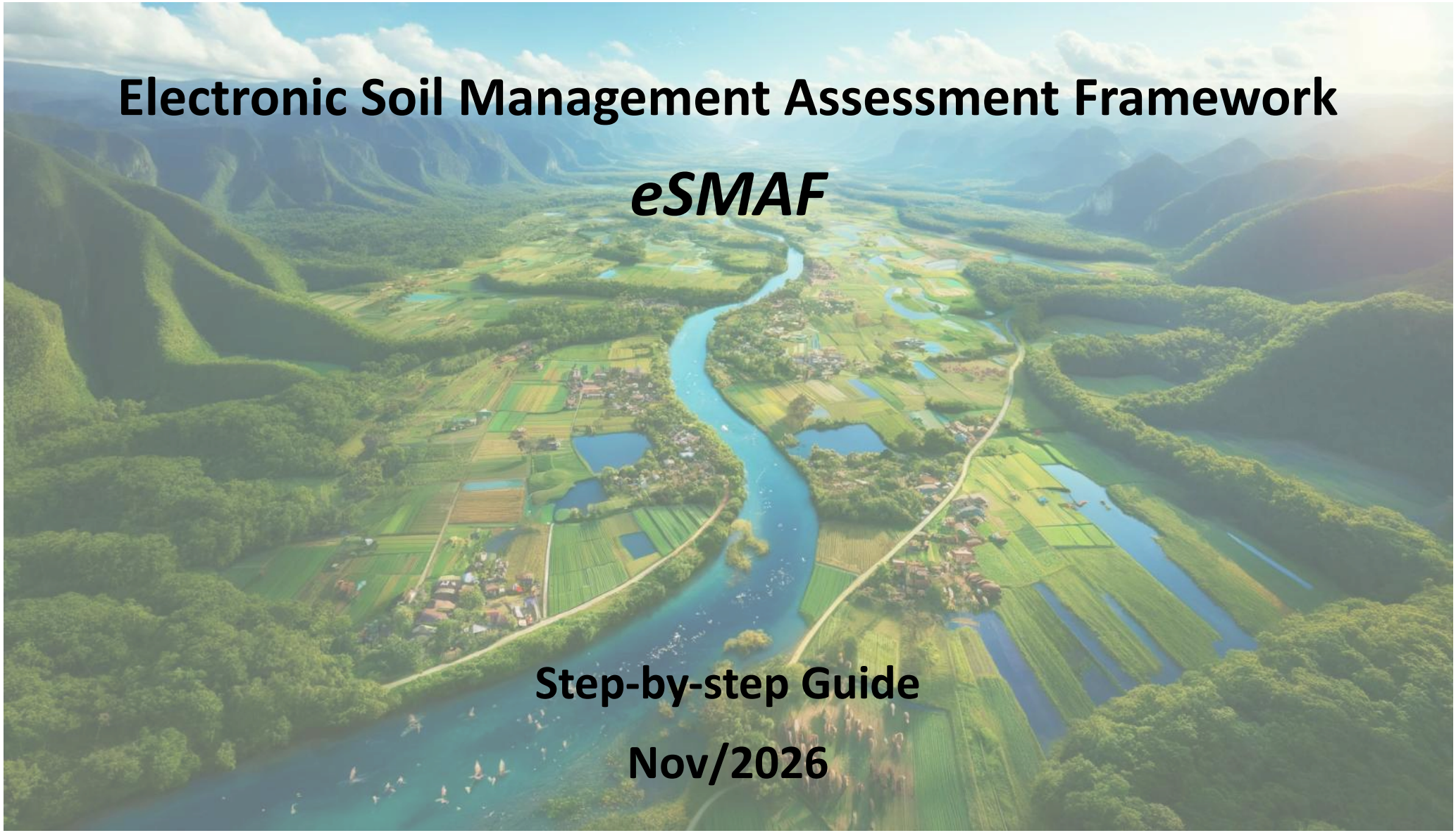
Electronic Soil Management Assessment Framework

eSMAF



Step-by-step Guide

Nov/2026

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Electronic Soil Management Assessment Framework (eSMAF)

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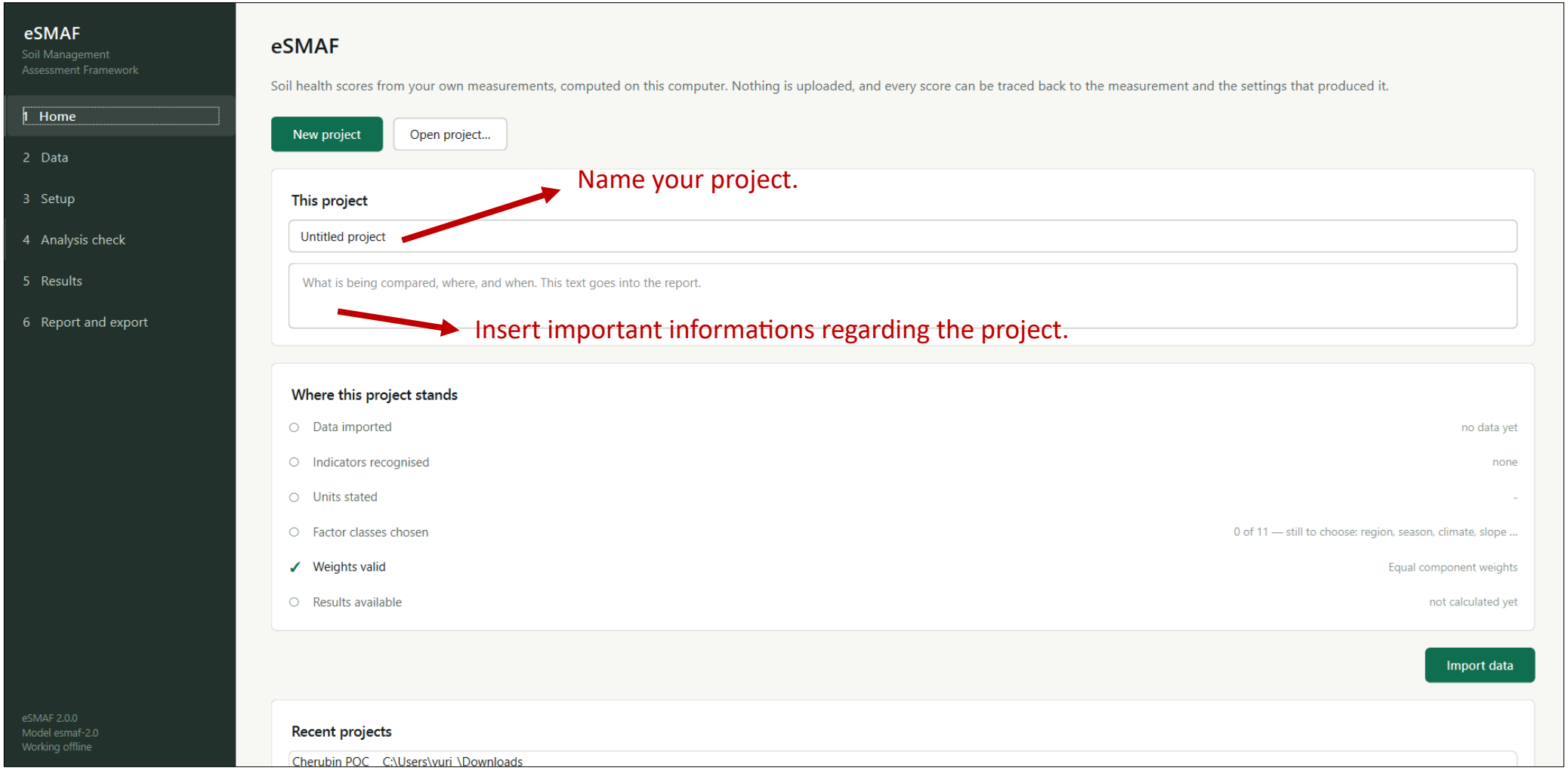
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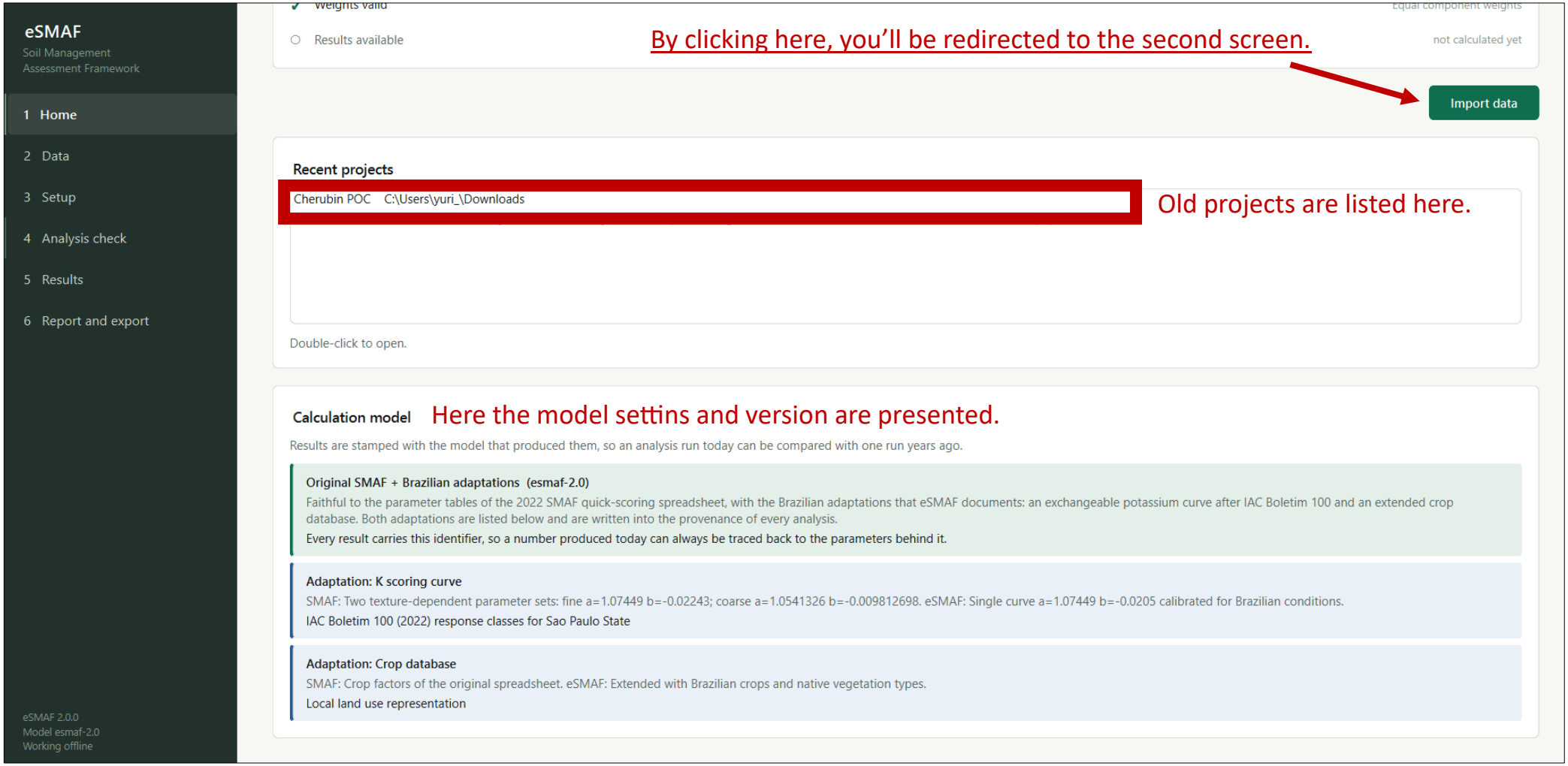
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1. Home

1.1. Top of the page

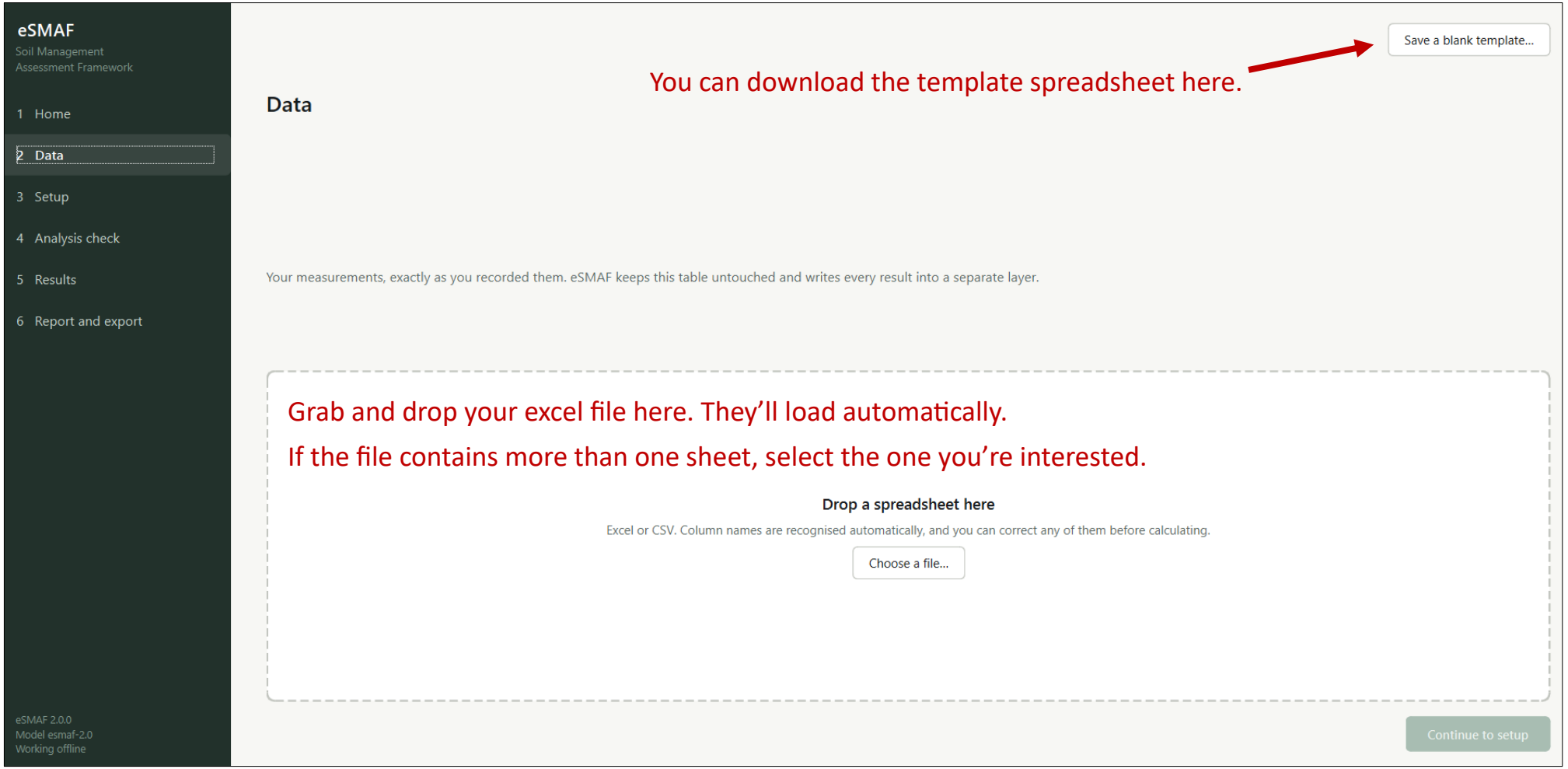


1.2. Bottom of the page



2. Data

2.1. Introducing your data



2.2. Template spreadsheet

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
1	site	land_use	management	crop	depth	sample	state	city	BD_g/cm3	AGG_pct	WFPS_fraction	AWC_gH2O/gSoil	pH_H2O	P_mg/kg	K_mg/kg	SAR	EC_dS/m	SOC_pct	PMN_mg/kg	MBC_mg/kg	BG_mg/kg/h
2									*g cm-3*	*%*	*fraction (0-1)*	*g H2O g-1 soil*	*-*	*mg kg-1*	*mg kg-1*	*(mmol L-1)^0.5*	*dS m-1*	*%*	*mg kg-1*	*mg kg-1*	*mg p-nitrophenol kg-1 h-1*
3																					
4	The template contains all possible indicators and their expected unit. You can leave blank the ones you're not using.																				
5																					
6																					
7																					

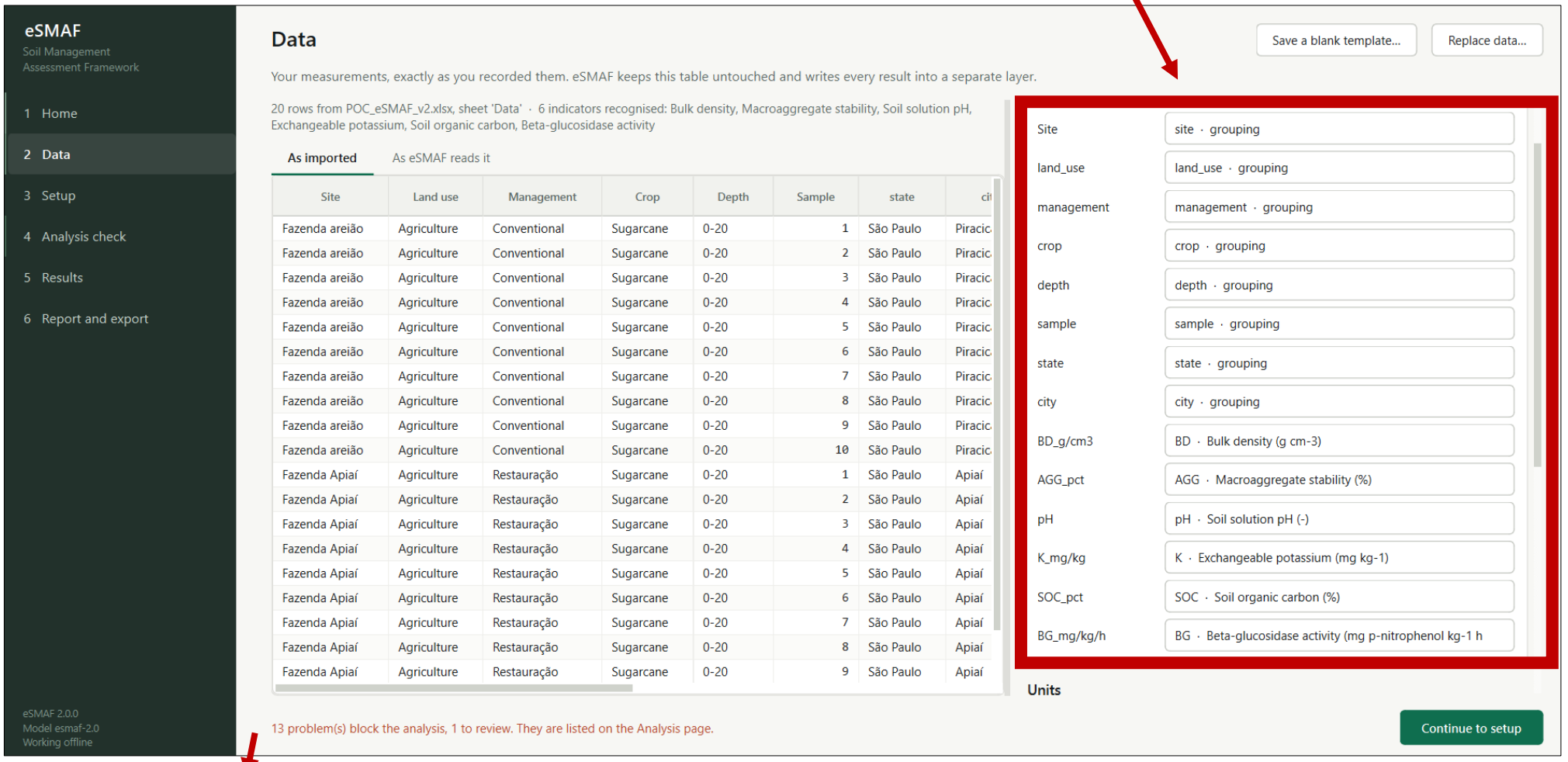
J11

	A	B	C	D
1	**Units eSMAF accepts**			
2	Write the unit into the column header, as the Data sheet does, and eSMAF will read it. Whatever the header says, the application asks you to confirm the unit before it scores anything, and records the conversion it performed with the result.	In the "units" sheet, you may find further information about the indicators and different accepted units.		
3				
4	**Indicator**	**Header in the template**	**Also accepted**	**What eSMAF does with it**
5	Bulk density	BD_g/cm3	g cm-3	The basis the scoring curve is defined on.
6			kg dm-3	Numerically the same as g cm-3.
7	Macroaggregate stability	AGG_pct	%	The basis the scoring curve is defined on.
8			fraction (0-1)	Multiplied by 100.
9	Water-filled pore space	WFPS_fraction	fraction (0-1)	The basis the scoring curve is defined on.
10			%	Divided by 100.
11	Available water capacity	AWC_gH2O/gSoil	g H2O g-1 soil (gravimetric)	The basis the scoring curve is defined on.
12			cm3 cm-3 (volumetric)	Divided by the FIELD bulk density of the same sample, which the dataset must carry: volumetric water content refers to the volume of intact soil, not of a sieved
13	Soil solution pH	pH_H2O	pH in water (1:1 or 1:2.5)	The basis the scoring curve is defined on.
14			pH in CaCl2 0.01 mol L-1	Reads below the water value; an offset is added.
15			pH in KCl 1 mol L-1	Reads below the water value; an offset is added.
16	Extractable phosphorus	P_mg/kg	mg kg-1	The basis the scoring curve is defined on.
17			mg dm-3 (per volume of sieved soil)	Divided by the density of the sieved sample.
18	Exchangeable potassium	K_mg/kg	mg kg-1	The basis the scoring curve is defined on.
19			mg dm-3 (per volume of sieved soil)	Divided by the density of the sieved sample.
20			mmolc dm-3 (charge, per volume)	Multiplied by 39.1 to reach mg dm-3, then divided by the density of the sieved
21			cmolc dm-3 (charge, per volume)	Multiplied by 391 to reach mg dm-3, then divided by the density of the sieved
22	Sodium adsorption ratio	SAR	dimensionless ratio	The basis the scoring curve is defined on.
23	Electrical conductivity	EC_dS/m	dS m-1	The basis the scoring curve is defined on.
24			mS cm-1	Numerically the same as dS m-1.
25			uS cm-1	Divided by 1000.
26	Soil organic carbon	SOC_pct	organic carbon, %	The basis the scoring curve is defined on.
27			organic carbon, g kg-1	Divided by 10.
28			organic carbon, g dm-3	Divided by the density of the sieved sample, then by 10.
29			organic MATTER, %	Divided by 1.724 (Van Bemmelen) to estimate carbon.
30			organic MATTER, g kg-1	Divided by 1.724 and by 10.
31			organic MATTER, g dm-3	Divided by the density of the sieved sample, by 1.724 and by 10.
32	Potentially mineralizable nitrogen	PMN_mg/kg	mg kg-1	The basis the scoring curve is defined on.
33			mg dm-3	Divided by the density of the sieved sample.
34	Microbial biomass carbon	MBC_mg/kg	mg kg-1	The basis the scoring curve is defined on.
35			mg dm-3	Divided by the density of the sieved sample.
36	Beta-glucosidase activity	BG_mg/kg/h	mg PNP kg-1 h-1	The basis the scoring curve is defined on.
37			mg PNP dm-3 h-1	Divided by the density of the sieved sample.
38			umol PNP g-1 h-1	Multiplied by 139.11 to reach mg kg-1 h-1.
39				
40				

Data | Units

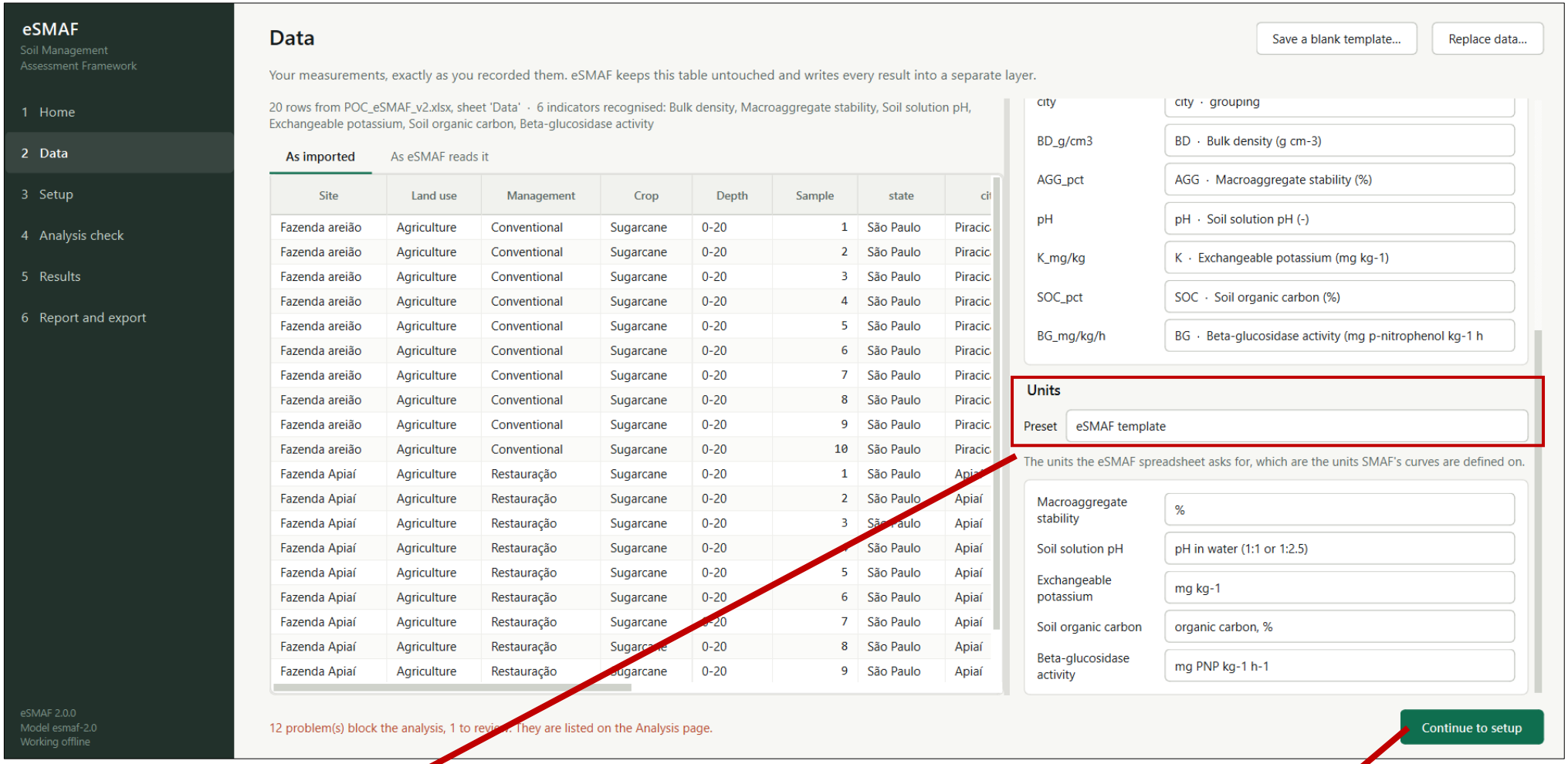
2.3. Data settings – individual classification

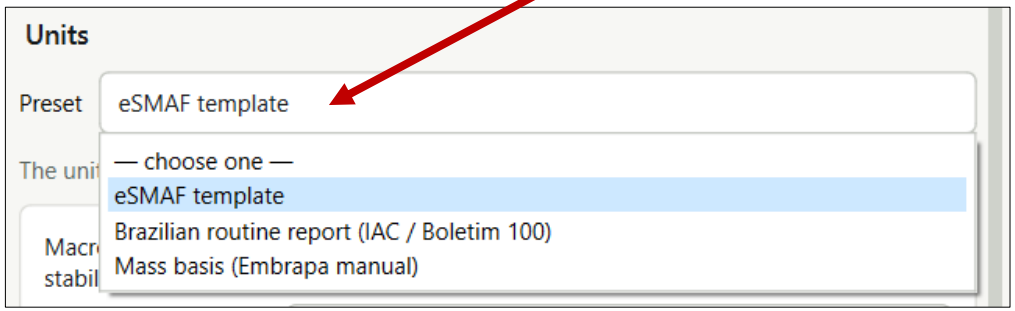
Confirm the grouping columns, indicators and units.



Here it indicates if there's any checking necessary after the software checked the data.

2.4. Data settings – predetermined classification



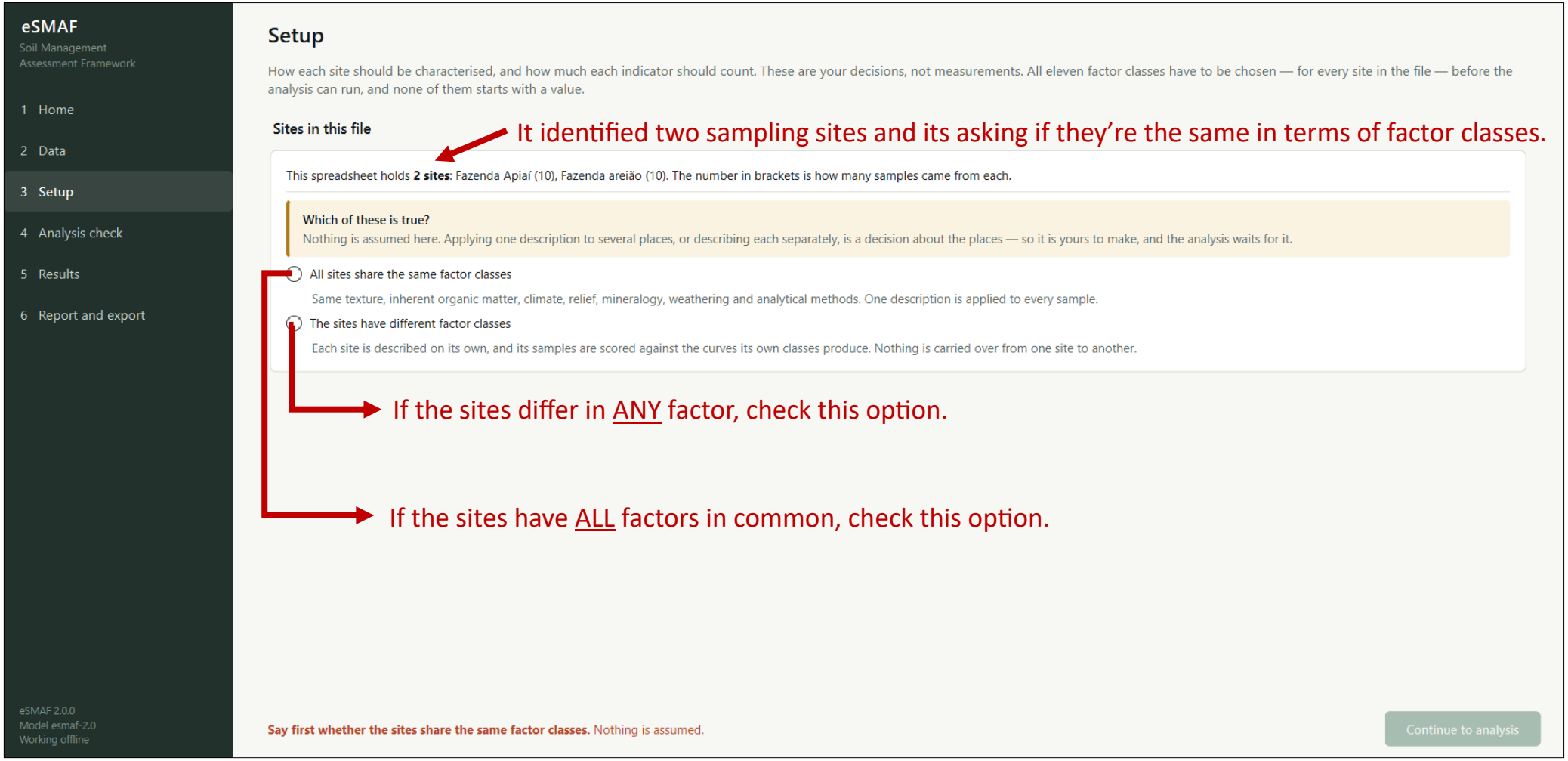


Clich here to go to the third screen.

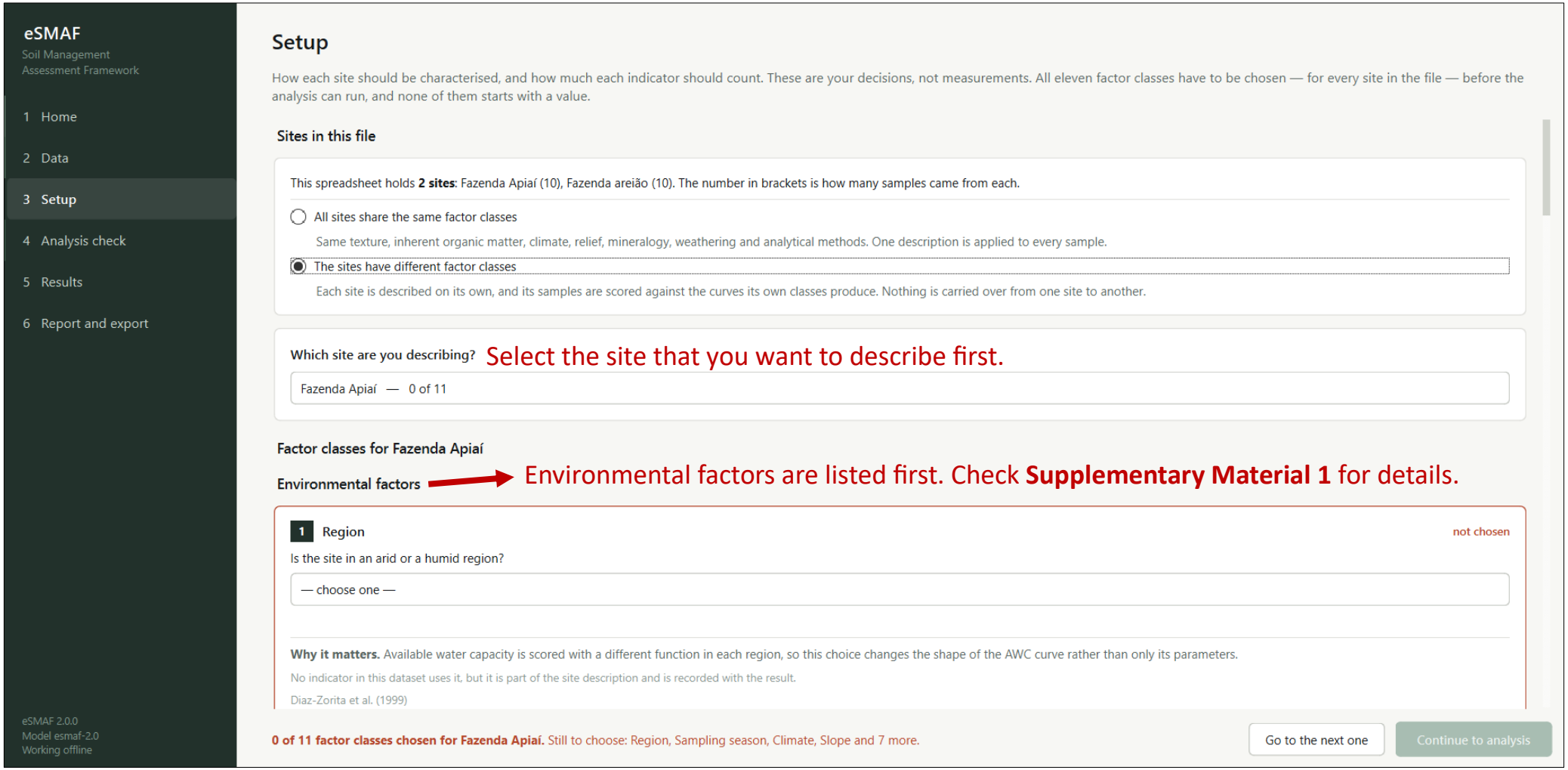
You can also select some type of preset units if you're using them directly from a soil analysis report.

3. Setup

3.1. Site determination



3.2. Factor classes inputs



EVERY FACTOR MUST BE CLASSIFIED!

3.2.1. Environmental Factors – *Region* and *Sampling Season*

Check **Supplementary Material 1** for details.

Factor classes for Fazenda Apiaí

Environmental factors

1 Region ✓

Is the site in an arid or a humid region?

2 — Humid region

Annual precipitation above the arid threshold (USDA LRR A, K-V).

Why it matters. Available water capacity is scored with a different function in each region, so this choice changes the shape of the AWC curve rather than only its parameters.

No indicator in this dataset uses it, but it is part of the site description and is recorded with the result.

Diaz-Zorita et al. (1999)

A brief explanation of “why it matters” and the reference is given.

2 Sampling season ✓

In which season, relative to the crop cycle, was the soil sampled?

1 — Spring - pre-planting or at planting

Reference season; no climate correction is applied.

Why it matters. Microbial biomass swings widely over a season. SMAF corrects for this by combining season with climate into a single factor, so that samples taken at different moments remain comparable.

No indicator in this dataset uses it, but it is part of the site description and is recorded with the result.

Franzluebbers et al. (1996)

EVERY FACTOR MUST BE CLASSIFIED!

3.2.2. Environmental Factors – *Climate* and *Slope*

Check **Supplementary Material 1** for details.

3 Climate ✓

How do sunny days and annual rainfall combine at this site?

1 — High / high

170 sunny days or more and 550 mm or more of annual precipitation.

Why it matters. Climate sets how much carbon and biological activity a soil can be expected to hold, and therefore what counts as a good score for the biological indicators.

Affects: Soil organic carbon, Beta-glucosidase activity

STATSGO / US Soil Survey

4 Slope ✓

What is the slope of the sampling site?

1 — 0 to less than 2 %

Why it matters. Slope sets the phosphorus concentration above which the environmental loss-risk limb of the P curve begins. On steep ground the penalty for excess P starts earlier.

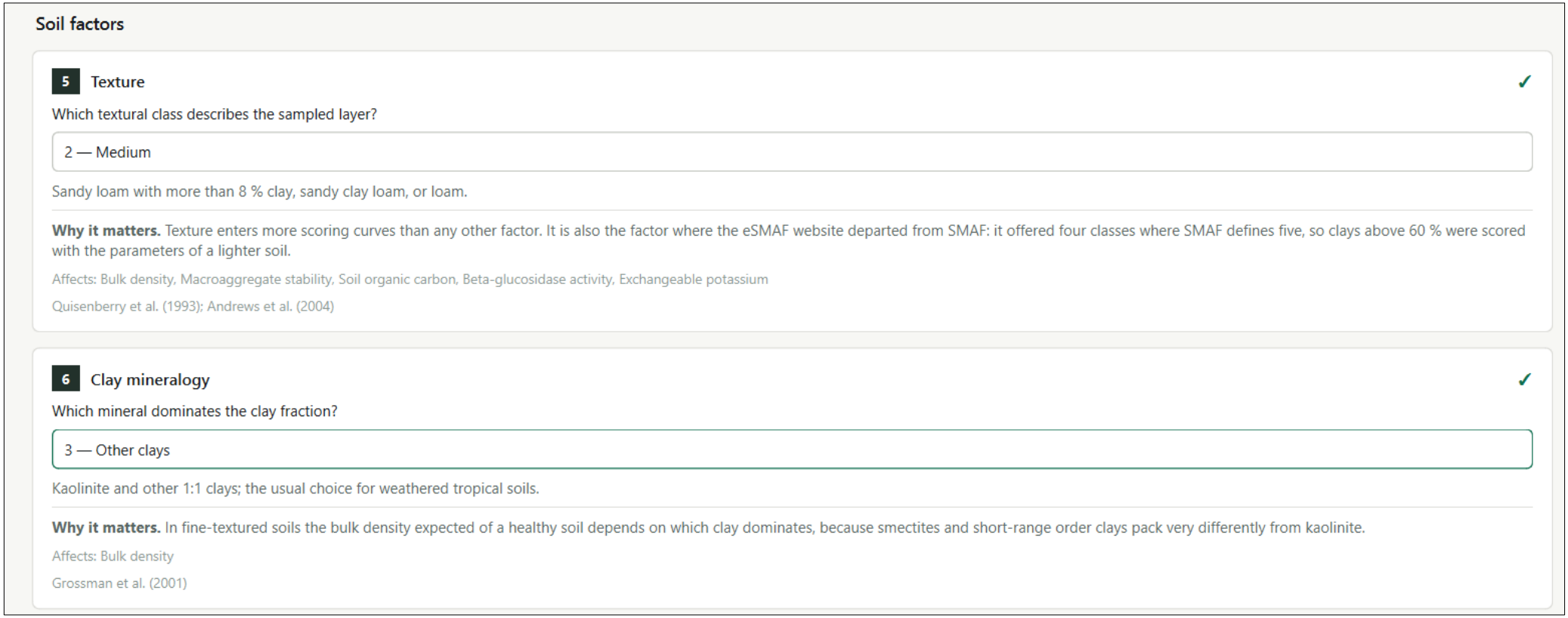
No indicator in this dataset uses it, but it is part of the site description and is recorded with the result.

Sharpley (1991)

EVERY FACTOR MUST BE CLASSIFIED!

3.2.3. Soil Factors – *Texture* and *Clay Mineralogy*

Check **Supplementary Material 1** for details.



EVERY FACTOR MUST BE CLASSIFIED!

3.2.4. Soil Factors – Organic Matter and Iron Oxide

Check **Supplementary Material 1** for details.

7 Inherent organic matter class ✓

What organic matter level is INHERENT to this soil type?

3 — Moderate

Soil Taxonomy: all Alfisols; Anthrepts; Entisols with aquic regime. SiBCS: Latossolos Amarelos, Vermelhos e Vermelho-amarelos; Argissolos; Cambissolos Háplicos; Nitossolos Vermelhos.

Why it matters. This is the single most influential factor in SMAF and the one most often misread. It does not describe the organic matter you measured; it describes the organic matter this soil type would hold in its natural condition, and it is read from the soil classification. Getting it wrong shifts every biological score.

Affects: Macroaggregate stability, Soil organic carbon, Beta-glucosidase activity

Soil Taxonomy; Embrapa (2013) SiBCS

8 Iron oxide class ✓

Does the soil have a textural B horizon with low base saturation?

2 — All other soil orders

Why it matters. Free iron oxides cement aggregates, so the aggregate stability expected of a healthy soil is higher in these soils.

Affects: Macroaggregate stability

Andrews et al. (2004)

EVERY FACTOR MUST BE CLASSIFIED!

3.2.5. Soil Factors – *Weathering* Check **Supplementary Material 1** for details.

9 **Weathering class** ✓

How weathered is the soil, and does it contain carbonates?

2 — Highly weathered

1:1 clays with oxides and hydroxides of Fe and Al: Ultisols, Oxisols, acidic Ochrepts. The usual choice for Latossolos.

Why it matters. Weathering governs how much of the phosphorus recovered by an extractant is actually available, and so acts together with the extraction method on the P method factor.

No indicator in this dataset uses it, but it is part of the site description and is recorded with the result.

Sharpley (1991), SSSAJ 55:1038

3.2.6. Method Factors – Phosphorus and Electrical Conductivity Check **Supplementary Material 1** for details.

Method factors

10 **Phosphorus extraction method** ✓

Which extractant produced the reported phosphorus values?

5 — Ion exchange resin

The routine method in Brazilian laboratories.

Why it matters. Extractants recover very different fractions of soil P. The measured value is multiplied by a factor that depends on the method AND on the weathering class before it reaches the scoring curve, so a mismatch here moves the P score more than a laboratory error would.

No indicator in this dataset uses it, but it is part of the site description and is recorded with the result.

Sharpley (1991); Andrews et al. (2004)

11 **Electrical conductivity method** ✓

How was electrical conductivity measured?

1 — Saturation paste

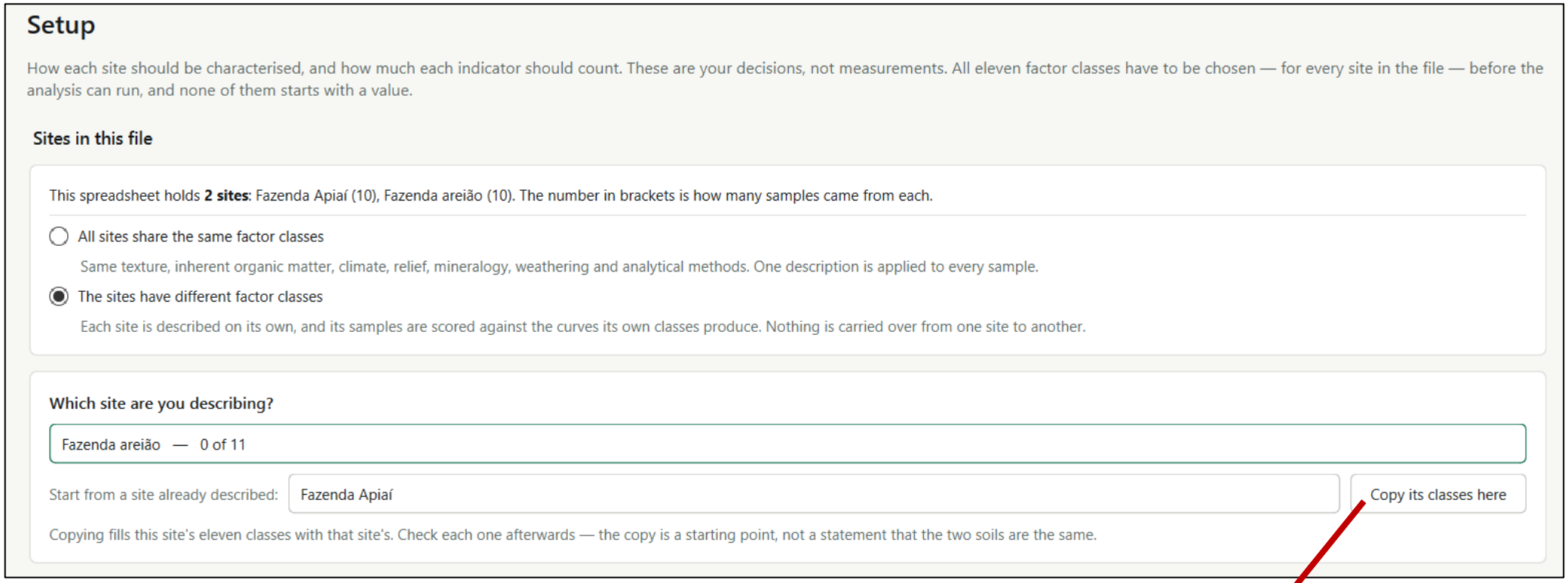
Why it matters. The salinity threshold of a crop is defined on a saturation paste. A 1:1 extract reads roughly 1.77 times lower, with a further texture correction, so the threshold has to be rescaled. The same conversion decides which branch of the SAR curve applies.

No indicator in this dataset uses it, but it is part of the site description and is recorded with the result.

Smith & Doran (1996)

EVERY FACTOR MUST BE CLASSIFIED!

3.3. Filling remaining site parameters

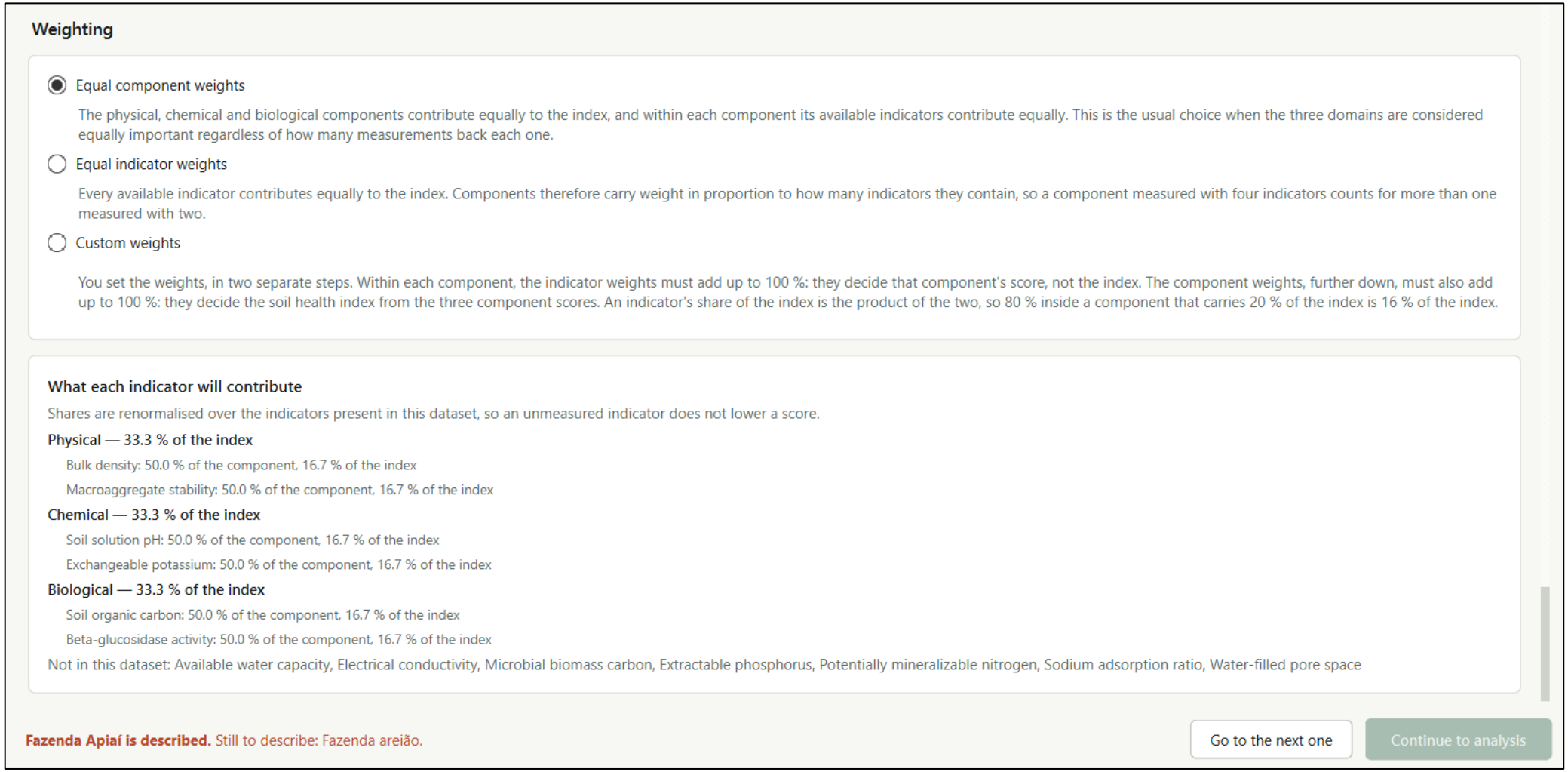


You can select this option to copy all factors to the new site and change only those necessary.

EVERY FACTOR MUST BE CLASSIFIED!

3.4. Setting the calculation mode

At the end of the page, you'll be presented 3 options of weighting. They're illustrated in the next page of this guide.



3.4.1. Three calculation modes

- EQUAL **COMPONENT** WEIGHTS



All components have the same weight in the index calculation.

- EQUAL **INDICATORS** WEIGHTS



The weight is divided equally between the number of indicators.
Components with more indicators have higher contribution in the final index.

- CUSTOM WEIGHTS



The weight of indicators and components is set as the user inputs.

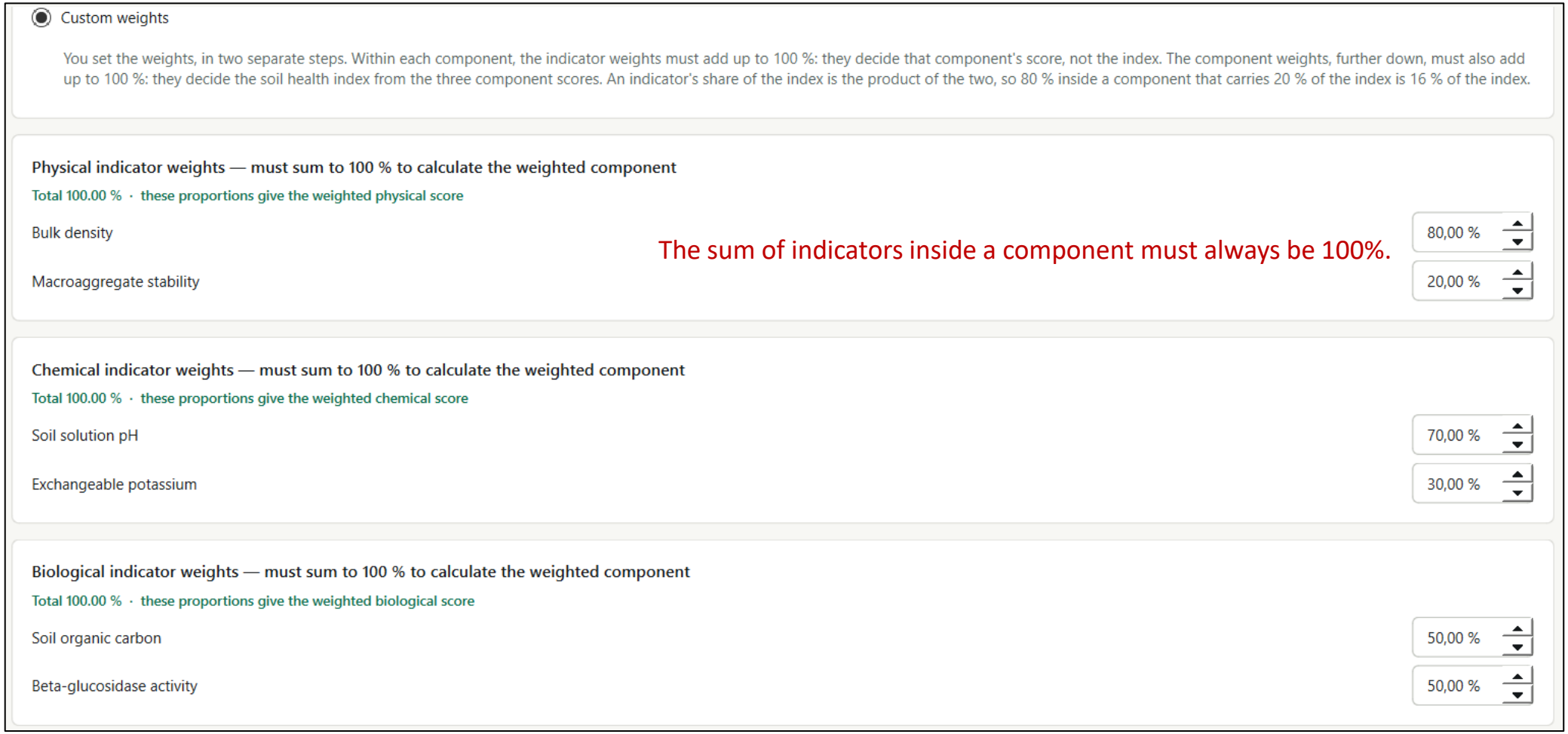
P = Phisical

B = Biological

C = Chemical

3.4.2. Custom Weights

Only if you choose **Custom weights** you'll have the following options to balance the weights as presented below.



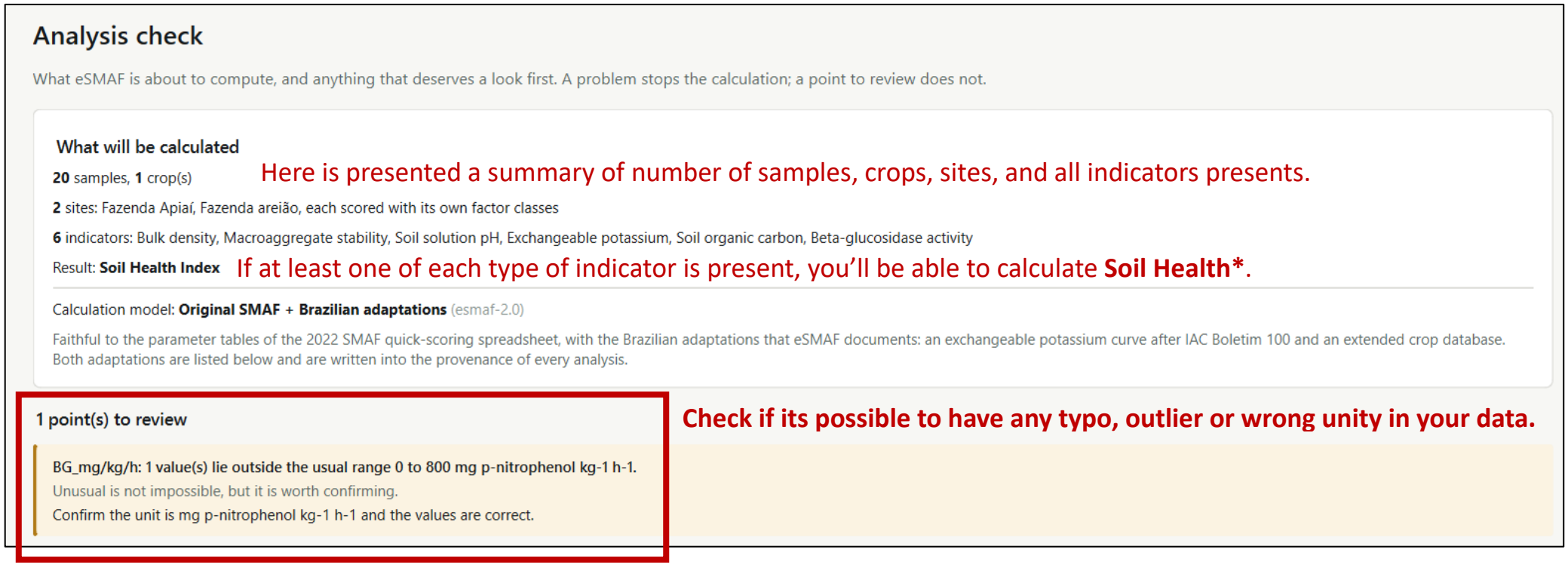
The sum of indicators inside a component must always be 100%.

You may also set the weights for each **Component** (not shown here).

They also need to sum 100%.

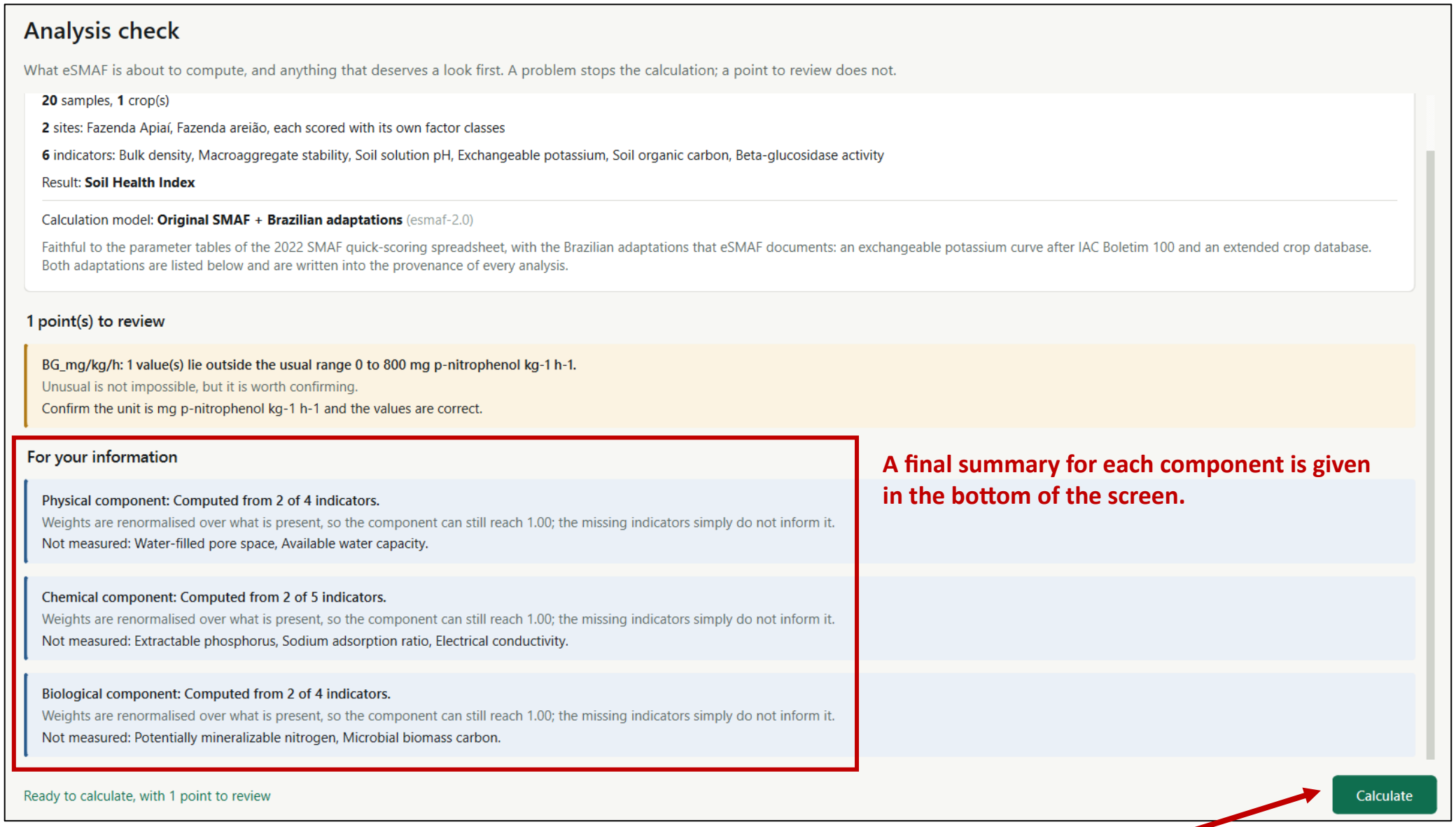
4. Analysis

4.1. Check summary



* Otherwise, you'll calculate the index for the combination of the components (Physico-Chemical Quality Index).

4.2. Points to review and indicators summary

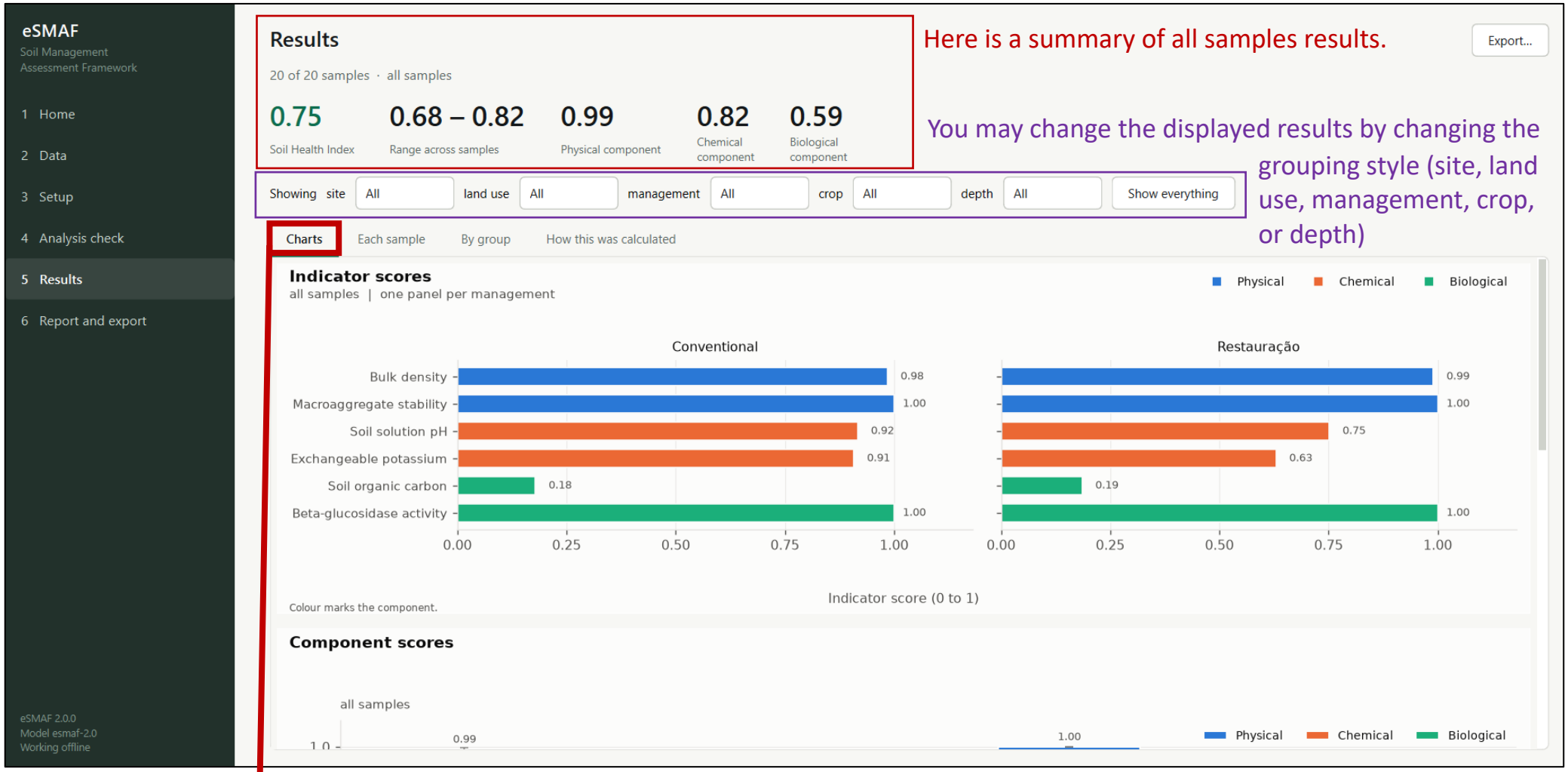


A final summary for each component is given in the bottom of the screen.

Click here to calculate!

5. Results

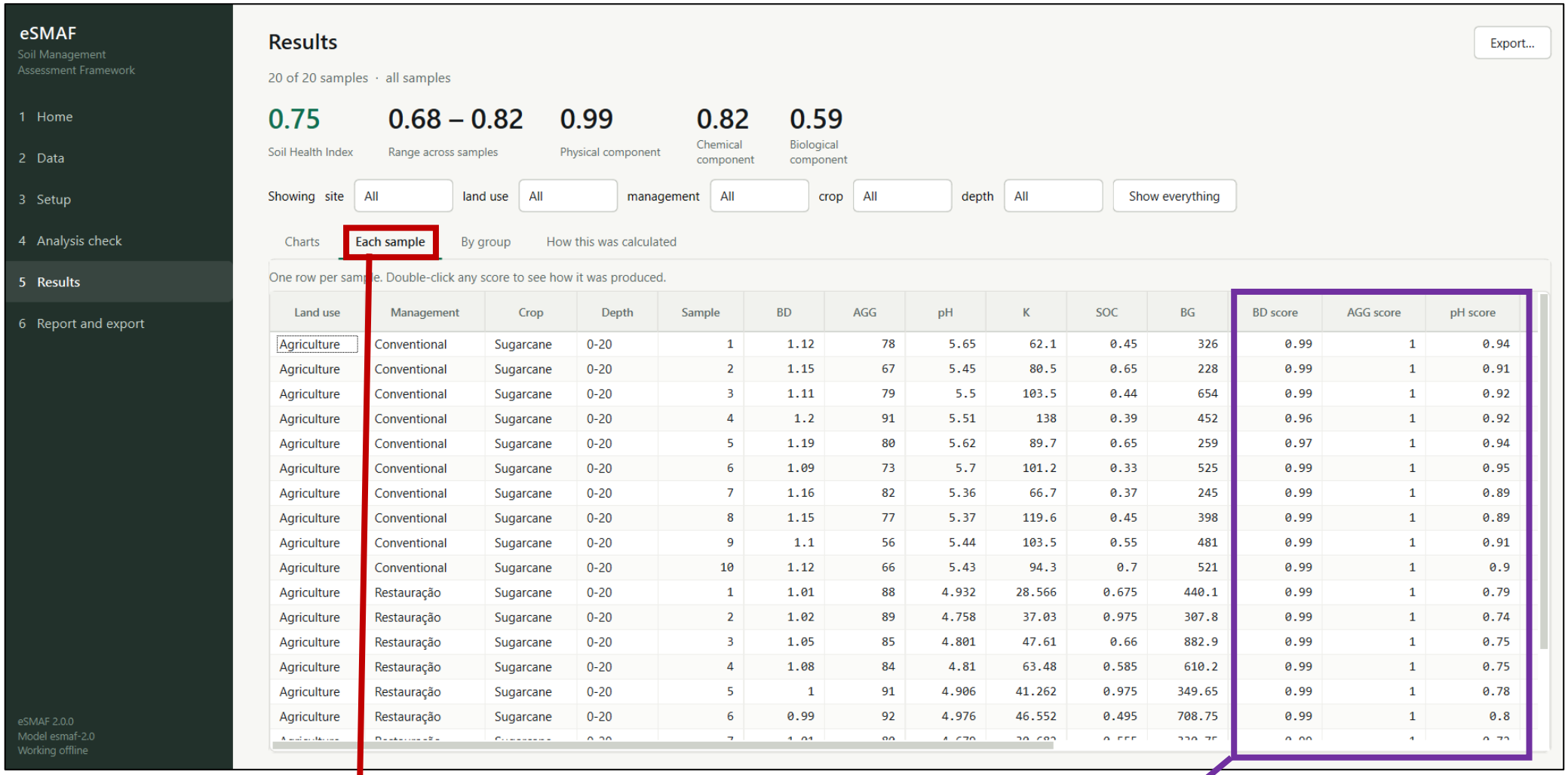
5.1. Main screen



The first tab presents the charts, and they will change according to the grouping style selected above.

The default is to present all samples, with no grouping.

5.2. Individual results (sample results)



Land use	Management	Crop	Depth	Sample	BD	AGG	pH	K	SOC	BG	BD score	AGG score	pH score
Agriculture	Conventional	Sugarcane	0-20	1	1.12	78	5.65	62.1	0.45	326	0.99	1	0.94
Agriculture	Conventional	Sugarcane	0-20	2	1.15	67	5.45	80.5	0.65	228	0.99	1	0.91
Agriculture	Conventional	Sugarcane	0-20	3	1.11	79	5.5	103.5	0.44	654	0.99	1	0.92
Agriculture	Conventional	Sugarcane	0-20	4	1.2	91	5.51	138	0.39	452	0.96	1	0.92
Agriculture	Conventional	Sugarcane	0-20	5	1.19	80	5.62	89.7	0.65	259	0.97	1	0.94
Agriculture	Conventional	Sugarcane	0-20	6	1.09	73	5.7	101.2	0.33	525	0.99	1	0.95
Agriculture	Conventional	Sugarcane	0-20	7	1.16	82	5.36	66.7	0.37	245	0.99	1	0.89
Agriculture	Conventional	Sugarcane	0-20	8	1.15	77	5.37	119.6	0.45	398	0.99	1	0.89
Agriculture	Conventional	Sugarcane	0-20	9	1.1	56	5.44	103.5	0.55	481	0.99	1	0.91
Agriculture	Conventional	Sugarcane	0-20	10	1.12	66	5.43	94.3	0.7	521	0.99	1	0.9
Agriculture	Restauração	Sugarcane	0-20	1	1.01	88	4.932	28.566	0.675	440.1	0.99	1	0.79
Agriculture	Restauração	Sugarcane	0-20	2	1.02	89	4.758	37.03	0.975	307.8	0.99	1	0.74
Agriculture	Restauração	Sugarcane	0-20	3	1.05	85	4.801	47.61	0.66	882.9	0.99	1	0.75
Agriculture	Restauração	Sugarcane	0-20	4	1.08	84	4.81	63.48	0.585	610.2	0.99	1	0.75
Agriculture	Restauração	Sugarcane	0-20	5	1	91	4.906	41.262	0.975	349.65	0.99	1	0.78
Agriculture	Restauração	Sugarcane	0-20	6	0.99	92	4.976	46.552	0.495	708.75	0.99	1	0.8

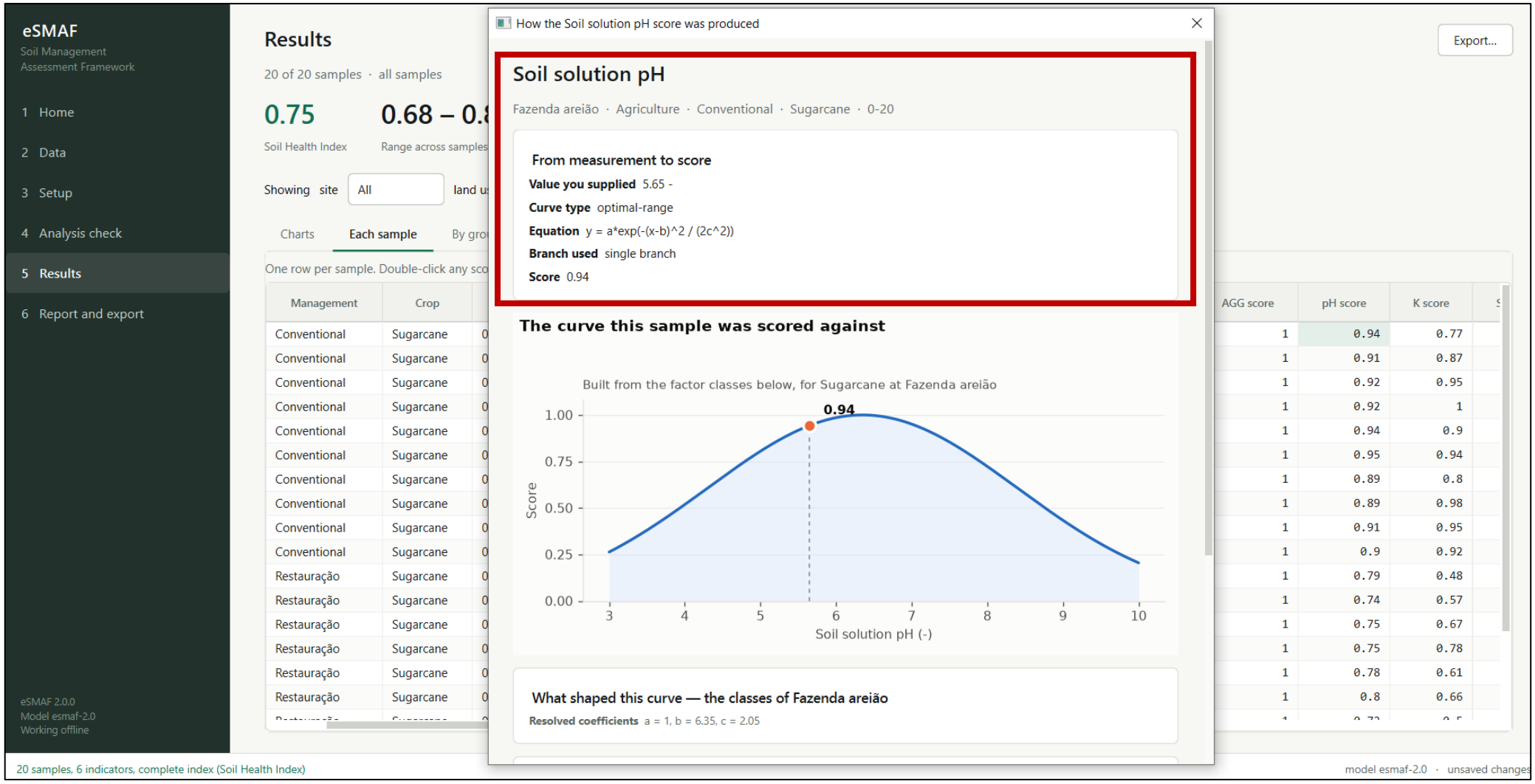
The second tab lists the results by individual samples. It can also be grouped in different styles.

Double-click any sample to open its curve details (presented in the next page).

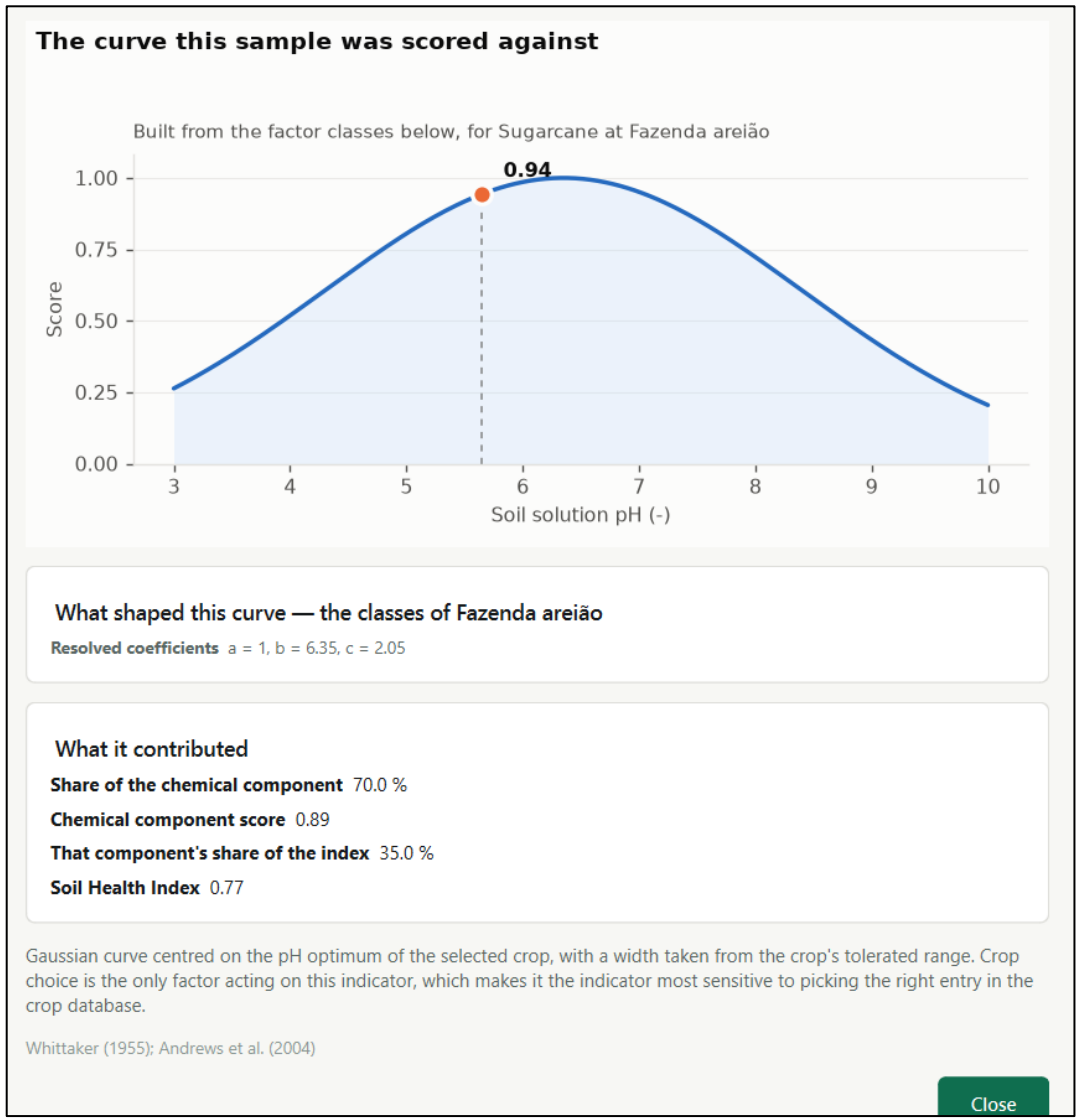
5.3. Score performance

5.3.1. Equation and type of curve

In the upper part, its presented the value entered (indicator value), the type of the curve, its equation and the score for that indicator.



5.3.2. Contribution details

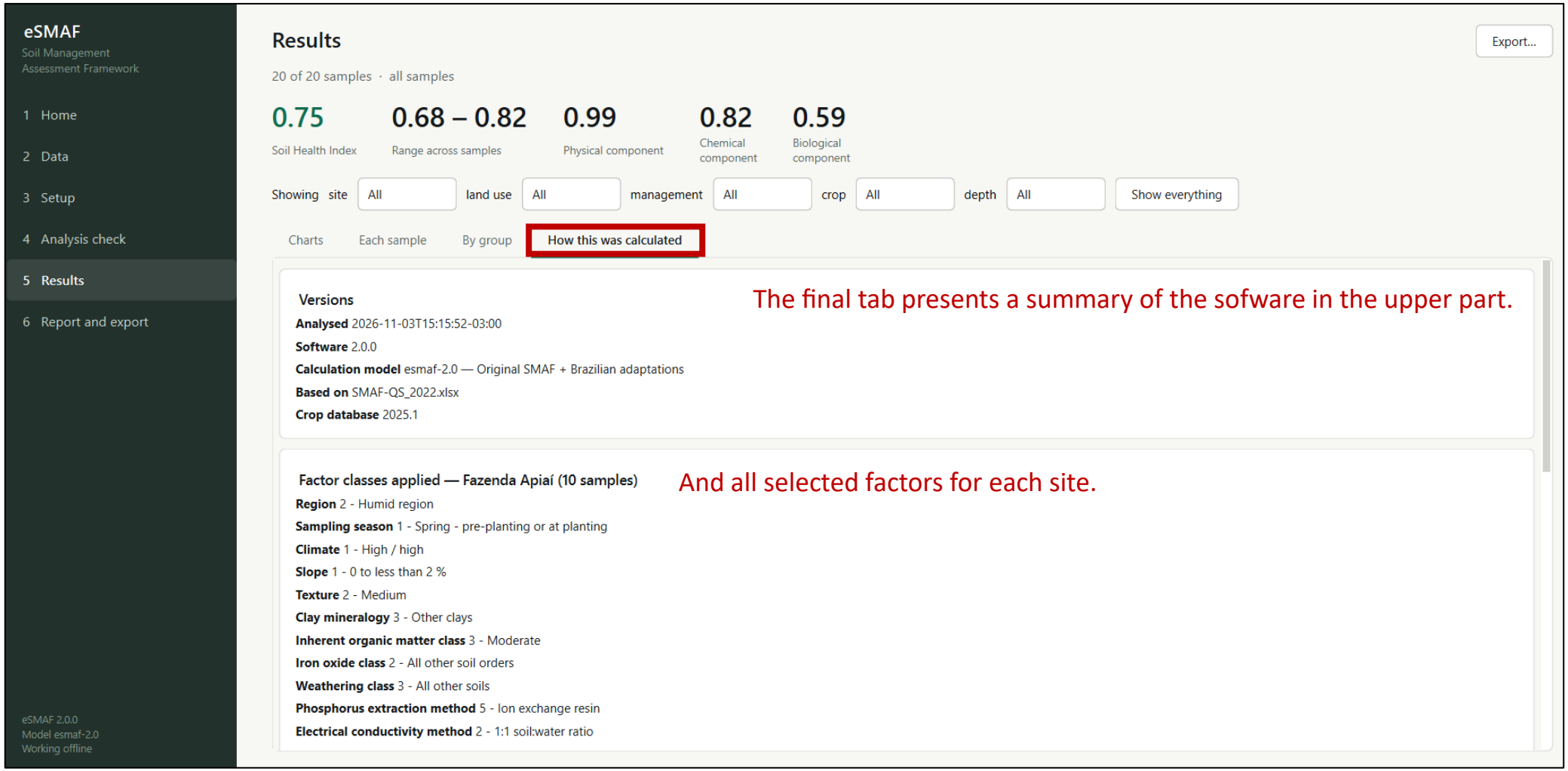


The graph presents where the analysed sample is located in reference to the curve calculated based on the factors entered previously.

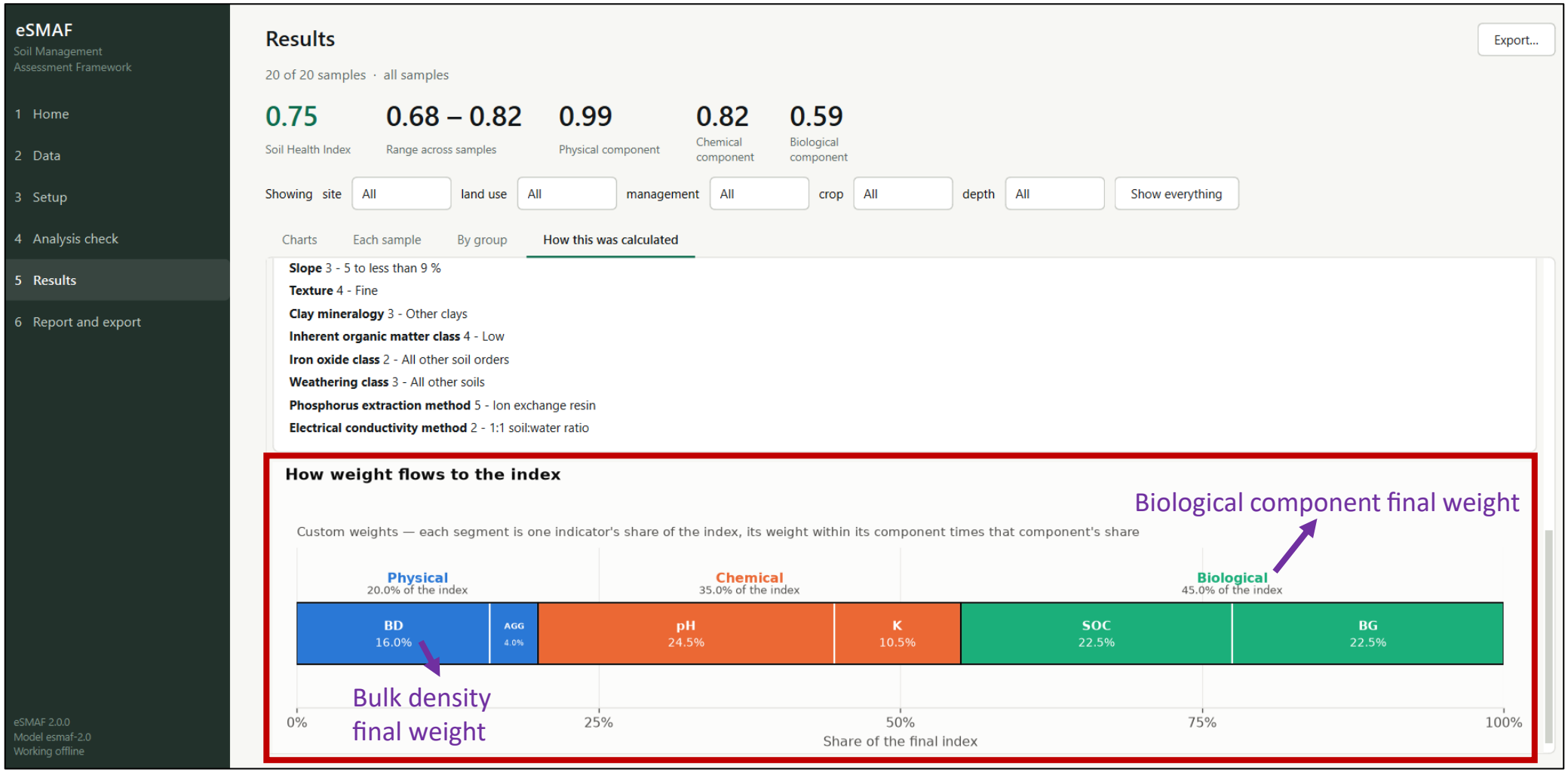
It also presents other informations about the weights and its participation in the index.

5.4. Parameters used for calculation

5.4.1. Version and factor classes



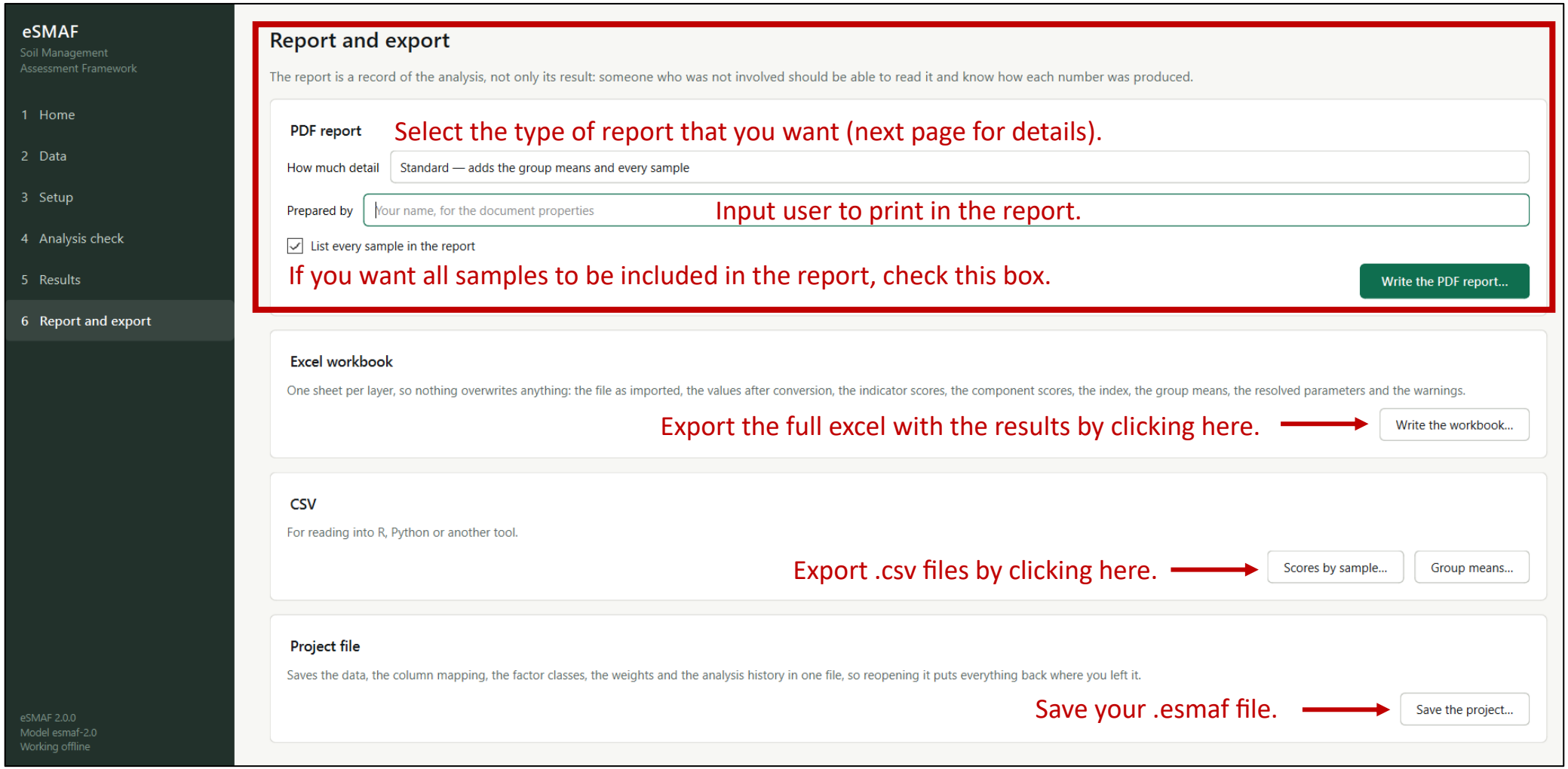
5.4.2. Indicators and components weights



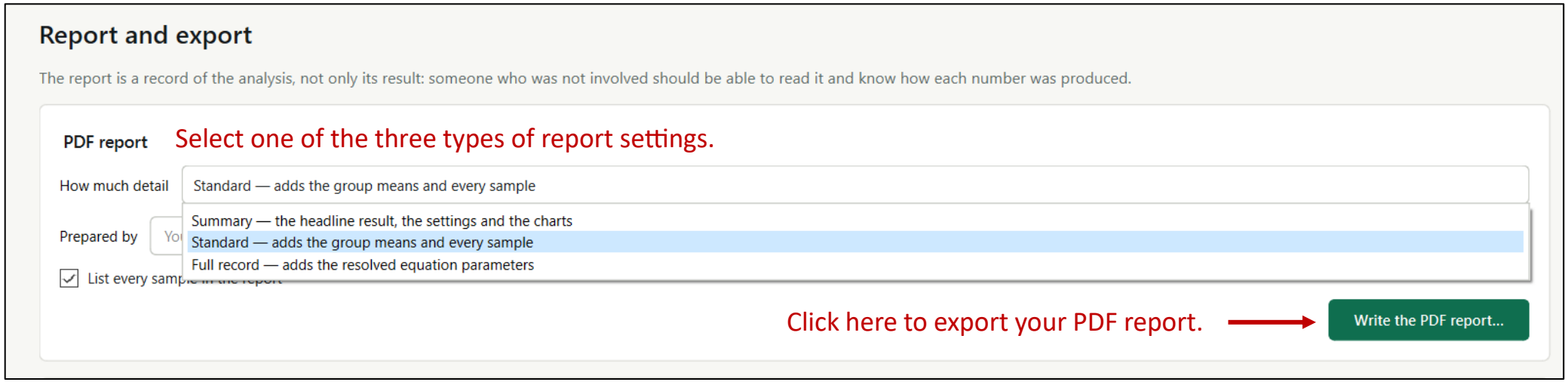
In the bottom, the index is presented as the indicators and components final weights.

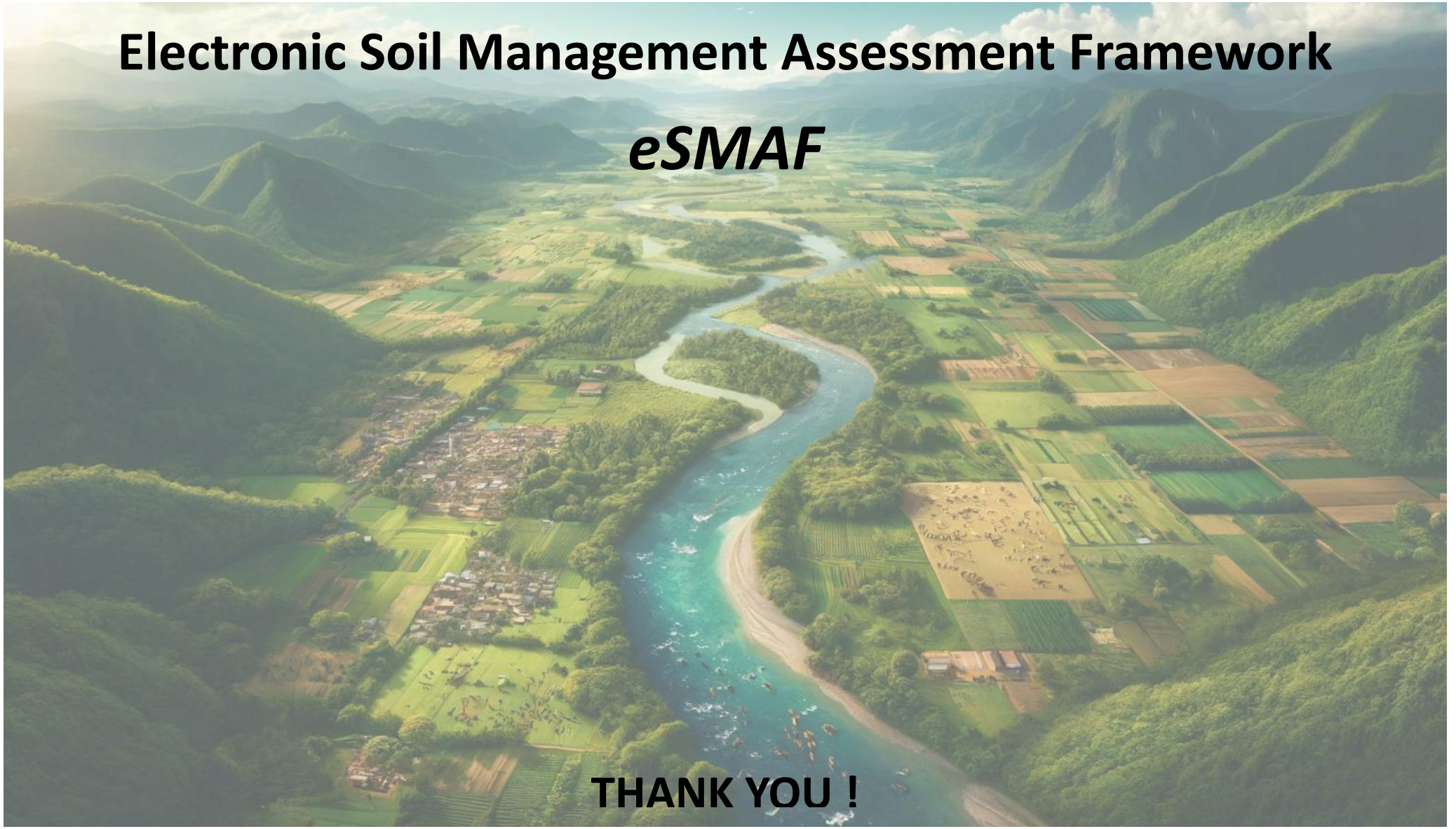
6. Export

6.1. Overview



6.2. Types of reports




Electronic Soil Management Assessment Framework
eSMAF
THANK YOU !